

DIVAAKAR NARESH

Backend Developer | AI/ML Engineer | Python Developer

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Chennai, India

Summary

AI/ML-focused Python Developer with hands-on experience building RAG pipelines, LLM applications, REST APIs, and backend systems. Experienced in FastAPI, LangChain, LangGraph, vector search, prompt engineering, document processing, and AI automation. Built RAG chatbots, computer vision systems, and business automation workflows.

Technical Skills

Languages: Python, Go, PHP, SQL **Backend:** FastAPI, REST APIs, Flask, Uvicorn
AI/ML: Machine Learning, Deep Learning, RAG, LangChain, LangGraph, LLMs, Prompt Engineering
AI Tools: Ollama, Qwen, MCP, Open WebUI, Chroma, Qdrant
Databases: PostgreSQL, Supabase, MySQL, MongoDB **DevOps:** AWS EC2, S3, Linux, Docker, Git, GitHub
Libraries: Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV, YOLO, ByteTrack

Experience

Freelance AI & Software Developer Chennai, India
Freelancer 09/2026 – 10/2026

- Built **RAG-powered chatbots** enabling users to query business-specific documents and knowledge bases.
- Developed **AI automation workflows** integrating LLMs, APIs, and backend services to automate repetitive tasks.
- Designed and deployed a **responsive website for a local business**.

Mavens i Softech Solutions Pvt Ltd Chennai, India
AI Engineer Intern 06/2026 – 08/2026

- Developed **CareShield**, an AI-driven health assessment platform with a FastAPI REST backend serving real-time LLM inference.
- Integrated **Qwen LLM** to generate persona-based, context-aware interview questions by analyzing user profiles and dynamically adapting conversations.
- Built an automated summarization pipeline converting collected responses into structured health summaries.
- Implemented prompt engineering and structured-output workflows for consistent and reliable LLM responses.
- Designed RESTful APIs for assessment workflows, response processing, and LLM inference using a modular architecture.

Sri Ramakrishna Math Chennai, India
Gen AI Intern 12/2025 – 01/2026

- Enhanced Open WebUI with organization-specific knowledge base capabilities for querying large document collections using LLM-powered retrieval.
- Implemented RAG pipelines using embeddings and vector similarity search for document-based question answering.

eNTrust Software & Services Pvt Ltd Chennai, India
IT Intern 06/2024 – 06/2024

- Explored Flask-based internal applications designed to optimize production workflows within a company network.
- Gained hands-on exposure to Python, web development, NLP, OCR, requirements gathering, and deployment practices.

Projects

Legal Document RAG System

- Built a legal RAG system using **FastAPI** and **Qdrant** for document-based question answering.
- Implemented structure-aware chunking based on articles, sections, and clauses.
- Stored document chunks with metadata to generate accurate, citation-backed responses.

AI-Powered Video Threat Detection System | Python, OpenCV, YOLO, ByteTrack, FastAPI

- Built a computer vision system analyzing surveillance footage to detect and track people and identify predefined security threats.
- Developed a frame-processing pipeline using **YOLO** for object detection and **ByteTrack** for persistent multi-object tracking.
- Implemented event-based detection for restricted-area entry and prolonged presence, generating timestamped alerts with confidence scores and tracking IDs.
- Developed a **FastAPI backend** for video upload, asynchronous analysis, result retrieval, and annotated footage generation.

Education

Ramakrishna Mission Vivekananda College Chennai, India
Bachelor of Computer Application – 8.02 CGPA 07/2023 – 05/2026

Santhome Higher Secondary School Chennai, India
12th Standard – 75% 06/2022 – 06/2023